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The Value of Social Media Data in Fashion Forecasting

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Abstract. *Problem definition:* How to use social media to predict style color and jeans fit sales for a retailer. *Academic/practical relevance:* Neither retail practice nor the academic literature provides a method for using social media to predict style color and jeans fit sales for a retailer. We present and validate a systematic approach for doing that. *Methodology:* Demand forecasting in the fashion industry is challenging due to short product lifetimes, long manufacturing lead times, and constant innovation of fashion products. We investigate the value of social media information for color trends and jeans fit forecasting. We partner with three multinational retailers, two apparel and one footwear, and combine their proprietary data sets with web-crawled publicly available data on Twitter and the Google Search Volume Index. We implement a variety of machine learning models to develop forecasts that can be used in setting the initial shipment quantity for an item, arguably the most important decision for fashion retailers. *Results:* Our findings show that fine-grained social media information has significant predictive power in forecasting color and fit demands months in advance of the sales season, and therefore greatly helps in making the initial shipment quantity decision. The predictive power of including social media features, measured by the improvement of the out-of-sample mean absolute deviation over current practice ranges from 24% to 57%. *Managerial implications:* To our knowledge, this study is the first to explore and demonstrate the value of social media information in fashion demand forecasting in a way that is practical and operable for fashion retailers. With consistent results across all three retailers, we demonstrate the robustness of our findings over market and geographic heterogeneity, and different forecast horizons. Moreover, we discuss potential mechanisms that might be driving this significant predictive power. Our results suggest that changes in fashion demand are driven more by “bottom-up” changes in consumer preferences than by “top-down” influence from the fashion industry.

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Keywords: retail operations • business analytics • empirical operations management

1. Introduction

Matching the supply of fashion footwear and apparel with demand is challenging due to a combination of factors. First, long manufacturing lead times combined with a short selling season means that most, if not all, of the supply for a season must be produced in advance of observing any sales data from the current season. For example, the three retailers we partnered with in this study had average lead times of 10 weeks, 14 weeks, and 12 weeks, respectively, and their initial shipment orders comprised 67.5%, 77.0%, and 77.7%, respectively, of eventual total season sales. A few “fast fashion” companies like Inditex are an exception to these long lead times, but the fact that they are viewed as outliers in the traditional fashion industry (Wallace 2016) shows that long lead times are the norm. Second,

most fashion products are newly developed for a particular season and have no prior sales data to guide a forecast of sales in the upcoming season. Finally, fashion products have short lifetimes, usually just a single season, so excess supply cannot be carried over for sales in the future. According to a Yale School of Management study (Baranda and Heck 2012) on the US branded retail apparel industry, both the industry average of days of inventory held and inventory levels in storage steadily increased over a five-year period, whereas out-of-stock rates have not reduced much. This indicates an increasingly difficult inventory management problem that hurts the revenue and gross margin of retailers.

Although many studies have been conducted across a variety of industries on predicting future demand based

on historical sales data, few have focused on the fashion industry. Kök and Fisher (2007) provide a comprehensive literature review of demand forecasting studies and find that those applied to the fashion industry failed to provide accurate and applicable predictions (Nenni et al. 2013).

As social networks take up a greater share of consumers’ time spent online, researchers are beginning to wonder if signals from social media could add value to the current practices for predicting various business outcomes. Indeed, recent increases in the quality and quantity of available social media data have already enabled social networks to be linked to user attributes and to business outcomes such as purchasing and response to advertising (Clemons et al. 2006, Moe and Trusov 2011). A recent Forbes article (Ward 2013) offers an anecdotal example that fashion retailers are beginning to use predictive analysis and data from social media and other sources to drive planning: “Reza Soudagar – Senior Director, Database and Technology Marketing at SAP ... cites an example of how sales personnel can make recommendations using predictive analytics fueled by Twitter posts, fashion industry trend data, a customer’s individual buying history, and even their fashion phobias.”

These trends motivated us to explore whether social media might be useful in forecasting demand for fashion footwear and apparel. We initially focused on forecasting color trends. For a specific product, retailers often offer a variety of color alternatives, so accurately predicting the demand for different color choices is important. The other reason to focus on color is that it is a defined and quantifiable fashion attribute. The most common color representation systems include the RGB (red, green, and blue) space and the LCH (lightness, chroma, and hue) space developed by the international commission on illumination (CIE). There is a one-to-one mapping between the two spaces. Color trend

refers to the change over time in percentage sales of different colors.

In our color study, we used social media related to color to improve forecasts of products at the style color level sold by three multinational retailers (two apparel and one footwear) that agreed to partner with us in this research. We used two streams of input data: transaction-level historical sales data from the three retailers and social media data related to fashion and color. The transaction-level sales data include, for each stock-keeping unit (SKU), its style description, size and color, category and subcategory, sticker price, transaction price, promotion information, inventory information, unit sales, and dollar sales.

Our social media data are comprised of tweets from 1,308 fashion bloggers posted between May 2013 and November 2017, their associated retweets and favorites, and the Google Search Volume Index (SVI) for apparel. The tweets came from fashion brands (our three focal retailers comprise three of the bloggers in our sample of 1,308), fashion magazines and other industry insiders, and individuals who comment on fashion and have attracted a following. A representative sample of sources is provided in Table 1. As of December 2019, Twitter is the second largest social media site after Facebook (Statista 2019). We choose tweets as our data source because Twitter emphasizes sharing ideas and information, whereas Facebook emphasizes connecting. Moreover, industry surveys (Ward 2013) report that shoppers rely on Twitter for information and advice, whether they are learning about a new product or on the brink of buying. More specifically, as Crum (2015) summarized, “Shoppers on Twitter are also engaged: they are 160% more likely to stay up to date on brand news and promotions, 120% more likely to search for deals and sales and 240% more likely to converse with a brand than retail shoppers on the average social network.” Information on

Table 1. A Sample of Fashion Influencers on Twitter

Twitter handler	Name	Category	Location	Number of followers ^a
@hm	H&M	Brand		3,423,441
@Burberry	Burberry	Brand	London	2,877,564
@TOMS	TOMS	Brand	Los Angeles	2,173,273
@MichaelKors	Michael Kors	Brand	New York	1,969,214
@CalvinKlein	Calvin Klein	Brand	New York	1,779,172
@Topshop	Topshop	Brand	London	847,299
@TommyHilfiger	Tommy Hilfiger	Brand	Global	549,446
@Coach	Coach, Inc.	Brand	New York	547,402
@tyrabanks	Tyra Banks	Celebrity		11,253,515
@nicolerichie	Nicole Richie	Designer		4,374,202
@bryanboy	Bryanboy	Fashion blogger	New York, London, Milan, Paris	583,980
@AIMEESONG	Aimee Song	Fashion blogger	Los Angeles	54,649
@wendynguyen	Wendy Nguyen	Fashion blogger	Los Angeles, CA	48,394
@tuulavintage	Jessica Stein	Fashion blogger	Sydney	34,720
@InStyle	InStyle	Fashion magazine	New York, NY	2,701,329
@MirandaKerr	Miranda Kerr	Model		2,785,791
@NETAPORTER	NET-A-PORTER.COM	Online fashion media	London	616,682

^aNote that the number of followers is the number at the time of web crawling.

Twitter was obtained by web-crawling techniques using Twitter's APIs (Application Program Interfaces).

We illustrate how color trends change over time and appear to be hard to predict with two sample styles in the sales data of one of the retailers. Figure 1 plots the percentage of unit sales split by color alternatives of two styles, each offered in four different color shades over time. Both styles offer the color gold but in different seasons, with the style on the top available from March to June 2014, and the style on the bottom available from January to March 2015. Note that a value of zero percentage of unit sales near the end of the timeline indicates a stock-out, and sudden surges near the end should not be interpreted as an increase in demand but rather are caused by other color alternatives being stocked-out. A remarkable finding is that the top graph suggests that gold was not the trend in 2014, whereas gold appears to be very popular in the bottom graph (2015) and sells out in less than two months.

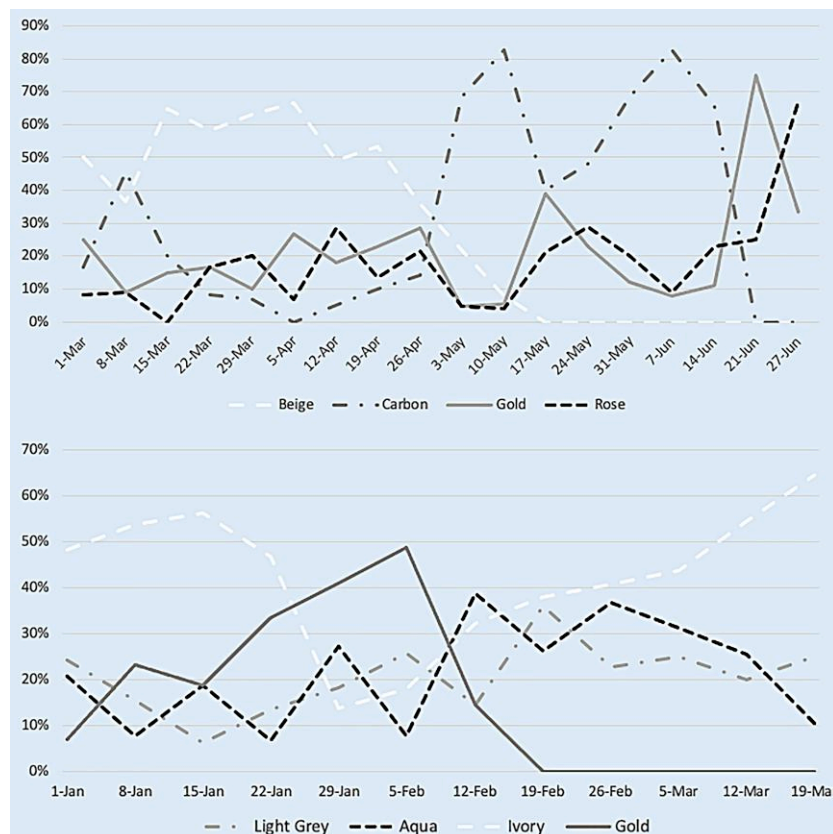
Figure 1 not only illustrates the existence of changing color trends but also demonstrates the difficulty of forecasting such demand. The beige alternative in the top graph and the gold alternative in the bottom were probably understocked, whereas some of the other alternatives were probably overstocked. The difficulty of forecasting at the time of the initial shipment quantity

for fashion retailers is well documented in the literature. For example, Fisher et al. (2001) reported that some women's styles can outsell the original forecast by 200%, and the average forecast error without early sales data can range from 60% to 127% (Fisher and Raman 1996). Recently, Ferreira et al. (2015) reported that the median demand forecasting error at a fashion retailer is around 50%.

Other defined fashion attributes include women's skirt length (one that is frequently commented on, including speculation that skirt length might correlate with business cycles), heel height in women's shoes, and jeans fit type. After completing our color study, we wanted to see how our method works for other attributes. This led us to investigate using social media for predicting jeans fit (with fit attribute levels Skinny, Slim, Straight, Girlfriend [GF], Boyfriend [BF], and Boot Flare) for one of our apparel retailers for whom this was an attribute and on which we had the necessary attribute and sales data. In this study, we used tweets related to bottoms and containing the above style keywords. These are the only two attributes we studied. Results for both are reported here.

This study has three main contributions. First, we apply natural language processing and machine learning techniques to extract fine-grained information from

Figure 1. (Color online) Percentage of Unit Sales of Two Styles by Colors



high-dimensional, coarse social media messages and use this to predict sales at the style color and style fit levels as much as six months in advance of sales.

Second, we validate our findings at three multinational fashion retailers in multiple geographic markets and across different supply chain lead times. We find that including social media information significantly improves demand forecast accuracy for these retailers. Our proposed forecasting method is also practical for most fashion retailers, as we forecast months in advance of the target season to cope with their long supply chain lead times. We also find significant heterogeneity across different geographic markets, indicating that fashion trends may have different diffusion patterns in different markets and that social media data are able to differentiate them.

Finally, we discuss why social media is predictive. By comparing which social media outlets are most predictive—top-down tweets from fashion companies as a form of advertising or bottom-up retweets, favorites, and Google SVI as an indication of consumer sentiment—we offer evidence on what drives fashion change, top-down industry push or bottom-up changes in consumer taste.

This paper is organized as follows. Section 2 provides a review of related research. Section 3 explains why social media might be useful by introducing the theoretical foundation of our approach and using this to develop our hypotheses. Section 4 describes details of the data sets used in the study, and Section 5 describes the forecasting framework that we developed based on theories and several forecasting models. Section 6 provides forecasting results. Section 7 investigates the impact of features, forecast horizon, and markets (Europe vs. United States). We validate our hypotheses and discuss mechanisms underlying our findings in Section 8. Finally, Section 9 concludes the paper with a summary of our results, insights obtained, and potential areas for future work.

2. Literature Review

Several papers in the operations management literature provide analytical models for managing inventory of products with short life cycles. See Raman (1999) for a thorough survey of this literature. Fisher and Raman (1996) and Fisher et al. (2001) show that early sales data can significantly improve forecast accuracy. Fisher and Vaidyanathan (2014) propose a consumer choice model in the product attribute space to forecast demand, especially for new products, and implement it at various retailers. The study conducted by Ferreira et al. (2015) is among the first to take a machine learning approach to demand forecasting at a fashion retailer.

Recently, Allon and Zhang (2015) introduced a new modeling framework that incorporates social networks, providing theoretical validation of the usage of social media data, and a number of researchers have begun to mine various social media to predict particular business outcomes.

Several studies used the Google Search Volume Index (SVI) for particular terms as the social media predictor, reasoning that a step in a purchase process is to research potential products using Google. Geva et al. (2017) predicted car sales by brand one month out using SVI on the brand names, Vosen and Schmidt (2011) forecast U.S. consumer spending using Google searches on 54 terms related to consumption, and Wu and Brynjolfsson (2009) forecast the volume and price of housing sales based on searches on terms related to housing.

Other studies have viewed social media as a type of publicity. Stephen and Galak (2012) forecast activity on a microlending site using mentions of the site in traditional and social media, and Luo et al. (2013) forecast the equity value of consumer products companies using online customer ratings and mentions of the firms on web blogs.

Tweets are often issued by product firms as a form of advertising. Crum (2015) reports on studies that Twitter commissioned with two market research firms to survey consumers on how they use Twitter in the buying process, and to then advise product companies on how to craft their messages to most effectively drive sales of their products.

Some papers have examined the role of social media in connecting people, and how one person's actions can influence the actions of their connected neighbors. Goel and Goldstein (2013) define a social network with 132 million people and 719 million edges, where an edge exists if a pair of individuals have sent emails to each other. They examine three different actions that people in the network can take—purchase in a department store, participate in the Yahoo Fantasy sports football league, and respond to online ads—and find that a person's decision to take these actions is increased if their network neighbors took the action. Trusov et al. (2009) show that sign-ups for social media websites are increased by invitations from existing members, which they liken to traditional word-of-mouth advertising. Goldenberg et al. (2012) consider, in the search for user-generated content, the role of social networks defined in part by hyperlinks between personal webpages, as in Facebook and LinkedIn. Centola (2010) conducted a live experiment in which individuals were shown to be more likely to sign up for a fictitious website if their network neighbors had signed up. Christakis and Fowler (2007) study not an online network, but a traditional network in which people in a town are joined by social

connections such as marriage, friendship, or siblings. They found that a person was more likely to be obese if people in their network were obese.

We note that the goals of the above studies are to use social media signals to predict a business result, not to explain why that result happened, a distinction well described in Shmueli (2010).

The recent pioneering paper by Cui et al. (2017) is the closest to our work. They forecast total daily sales for an online retailer of men's apparel with a forecast lead time of one to seven days. Key inputs to their forecast model are recent past sales and the length and sentiment of comments that customers post to the company's Facebook page in reaction to advertisements placed by the company on their Facebook page. They reason that positive sentiment expressed by customers would impact sales, first by creating awareness of a product, and then by signaling to a potential buyer that their friends like the product, a form of online word-of-mouth advertising. They conclude by suggesting that future research is needed to forecast "sales by category or SKU-level sales" and "longer-term forecasts," which, of course, is exactly what our paper does. In addition to providing long-term, fashion forecasts, we seek to complement their results and contribute to this field in several respects. First, rather than focusing on a single retailer's Facebook page (which by nature is likely to have an impact on the particular retailers near-future sales, as a majority of the posts are promotions; Lee et al. 2018), our input data set is generated from a collection of more than a thousand fashion influencers collected over a 4.5-year period. Second, we extend the forecasting lead time from days to months, which means we do not have sales available as a predictive variable, but it is more practical for most fashion retailers. Our forecasting results also shed light on why social media data might have predictive power. Finally, we validate our findings at three multinational fashion retailers.

3. What Drives Fashion Changes and Why Might Social Media Be Predictive of These Changes?

To understand the nature of the fashion supply chain and how it is being impacted by social media, we reviewed relevant publications and interacted with managers at our partner retailers and industry leaders. The industry leaders included Duncan Painter, chief executive officer, and Carla Buzzi, global content officer and director of color at WGSN, and Helen Slaven, managing director at PlanetRetail. **WGSN is the world's leading online research and trend analysis and news service for the fashion and style industries. PlanetRetail is the world's leading provider of global retail intelligence. Pantone is a fashion forecast company focused**

explicitly on color, which asserts that "Pantone's Color Trend Forecasting publications give decision makers the insight needed to feel confident that they are creating product in the right color at the right time." (Pantone 2021). The importance of color is well illustrated by Pantone's announcement of their "color of the year" for 2022 (a shade of periwinkle blue called "Very Peri") being front page news in the *Wall Street Journal* (Janin 2021).

These companies constantly watch for signs and clues of future fashion trends, from the runways, the fashion shows, what's showing in stores, the streets, the celebrities, and social media sites. **Alongside these companies predicting fashion trends sit the fashion brands and institutions like Vogue magazine, which seek to influence (they might say "dictate") fashion trends. Taken together, these institutions represent the legacy system for predicting and influencing fashion.**

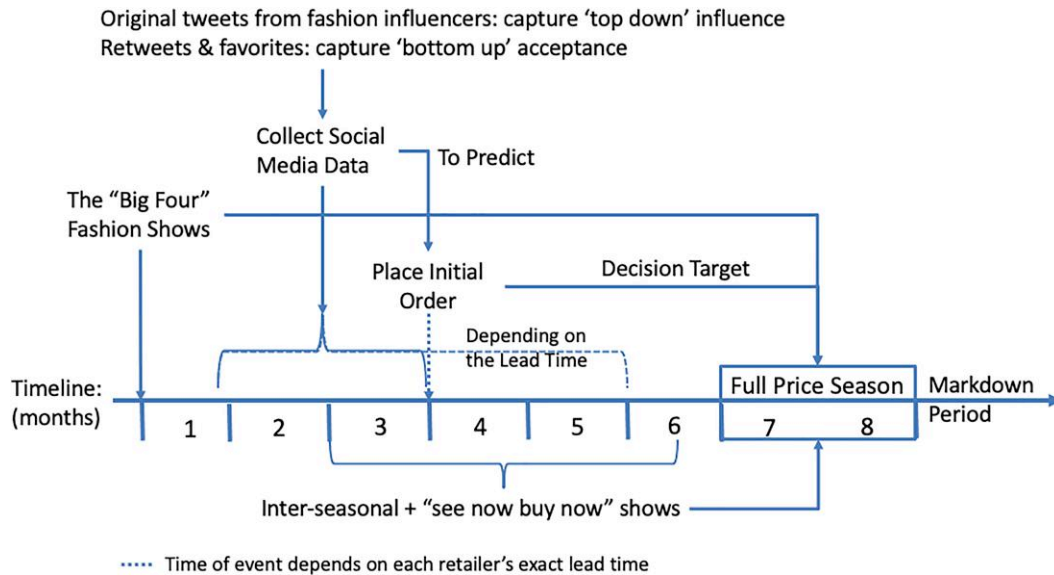
Social media provide a new channel through which this legacy system that has existed for decades can exert its influence. Fashion magazines are quickly turning to the digital space to provide print content (Kay 2017) and tweets from brands, designers, and fashion magazines as a new way for the industry to drive trends. Also, Google is launching a new program to allow brands themselves to control what consumers see first in search results, specifically around the time of prominent events like Fashion Week (Fateh 2017). **The market research firm Mintel released a report in 2015 in which they discovered that more than 35% of millennial women in America say social media is one of their top influencers when making clothing purchases (Mintel Press Team 2015).** Via social media platforms, brands can now have a direct relationship with their consumers.

Figure 2 depicts the typical North America-based fashion retailer planning calendar starting with fashion shows held well in advance of the selling season as a market survey and indicator, to ordering, manufacturing (mostly in Asia), shipping, and hitting the stores. All our subject retailers are based in North America, but they also have sales in Europe, where they follow the calendar in Figure 2.

According to Fashion Week Online (2016), the most renowned fashion weeks happen twice per year (February for fall/winter collections, and September for spring/summer collections) in the "Big Four" fashion capitals: New York, London, Milan, and Paris. Given this timeline, it is not surprising that advertisements of upcoming season styles and propaganda of upcoming fashion trends, including the tweets from bloggers that we analyze, start as early as those fashion shows. This leads us to believe that information from months ahead is related to the upcoming season demand.

We also learned about the nature of variation in color preferences over time. Black and white are always the two most popular colors, but secondary colors will wax and

Figure 2. (Color online) Fashion Retail Cycle of a Typical North America–Based Fashion Retailer and the Proposed Forecasting Approach



wane in popularity. Color shades also matter tremendously. The CEO of one of the apparel retailers admonished us at length that the question is not whether green will be popular but which exact shade of green. He emphasized that, in any season, some shades of green (or any other color) sold well and some sold poorly. And even white now comes in multiple shades.

Daniels (1951), Sproles (1981), and Miller et al. (1993) consider the fashion process and what drives changes in fashion preferences. They identify two competing forces driving fashion trends: (1) the fashion industry, comprised of brands, designers, and fashion magazines on the one hand, and (2) consumers on the other hand.

The tie-dyed T-shirts popular in the 1960s are a perfect example of a consumer-driven color fashion (really "antifashion," related to social protest and opposition to the Viet Nam war) trend. The fashion industry as a driver of trends is well depicted in the 2006 movie *The Devil Wears Prada*. The main characters are Miranda Priestly (played by Meryl Streep), the editor-in-chief of the number one fashion magazine, and her newly hired assistant Andrea Sachs (played by Anne Hathaway), a recent Columbia grad who disdains fashion (and was wearing a "lumpy blue sweater," a modern-day equivalent of the tie-dyed T shirt) but was working in the industry because she needed the job. In a memorable scene, Priestly uses not just a color, but a color shade, to lecture Sachs on how the fashion industry works.

Miranda Priestly: "You think this has nothing to do with you. You go to your closet, and you select out, oh I don't know, that lumpy blue sweater, for instance, because you're trying to tell the world that you take yourself too seriously to care about what you put on

your back. But what you don't know is that that sweater is not just blue, it's not turquoise, it's not lapis, it's actually cerulean. You're also blindly unaware of the fact that in 2002, Oscar de la Renta did a collection of cerulean gowns. And then I think it was Yves St Laurent, wasn't it, who showed cerulean military jackets? And then cerulean quickly showed up in the collections of eight different designers. Then it filtered down through the department stores and then trickled on down into some tragic 'casual corner' where you, no doubt, fished it out of some clearance bin. However, that blue represents millions of dollars and countless jobs and so it's sort of comical how you think that you've made a choice that exempts you from the fashion industry when, in fact, you're wearing the sweater that was selected for you by the people in this room."

Interviews with fashion insiders (Ross 2017) confirm that this movie accurately depicts how the industry push of fashion works, although, as noted by Sproles (1981, p. 118), it is uncertain how well it works. "Though anecdotes abound, there is surprisingly little empirical documentation of industry's role in propagating fashion trends."

Social media outlets not only provide an additional channel for top-down industry messages, they also empower bottom-up consumer sentiment, as noted by Imran Amed (founder and editor in chief of the *Business of Fashion*, an independent publication that has become to industry insiders what the *Financial Times* is to the public) in an interview with *The Guardian*: "In this new hierarchy, the consumer has the ability to amplify or negatively impact on business, through sharing positive or negative responses. Once brands and magazines

dictated what we should buy, now consumers are telling us what they like and want, and the power structure has been turned on its head” (Kay 2017).

Our social media data capture both the top-down industry push via tweets from industry insiders and the bottom-up consumer sentiment via retweets/favorites received by each tweet (consumer’s positive or negative responses) and Google Search Volume Index (SVI). The first is a type of industry advertising (and is likely correlated with other industry advertising efforts), and the second is an expression of consumer preference that functions the same way as advance purchase orders (Moe and Fader 2002), customer surveys (Jones 1975), and early sales data (Fisher et al. 2001). By comparing the relative impact of the top-down and bottom-up streams, we will provide evidence on the important question of *who drives fashion change—consumers wearing tie-dyed T-shirts and lumpy blue sweaters or Miranda Priestly and the designers introducing cerulean blue.*

Both top-down and bottom-up forces are likely to have an impact, which leads to our first hypothesis.

Hypothesis 1. *Social media data (tweets, retweets/favorites, and Google SVI) are predictive of apparel color and jeans fit demand in an upcoming season.*

The above observation (Kay 2017) of rising consumer power due to social media leads to our second hypothesis.

Hypothesis 2. *Retweets/favorites and Google SVI are more predictive than tweets of apparel color and jeans fit demand in an upcoming season.*

Many fashion companies are seeking to reduce their product development lead time in the belief that if they make design decisions closer to the start of the next season, then they will have better information on which to base those decisions. This leads to our third and final hypothesis.

Hypothesis 3. *The closer we are to the start of a season, the more predictive social media data (tweets, retweets/favorites, and Google SVI) will be of apparel color and jeans fit demand in an upcoming season.*

4. Data

We used two data sets to investigate these hypotheses: *tweets crawled from the social media site Twitter and the Google Search Volume Index (SVI) under the “Apparel” category*, and three proprietary transaction-level sales data sets from three multinational fashion retailers.

4.1. The Social Media Data

Ideally, we would collect tweets from every single Twitter account, but this is too massive to achieve using Twitter’s publicly available APIs. Instead, we compiled

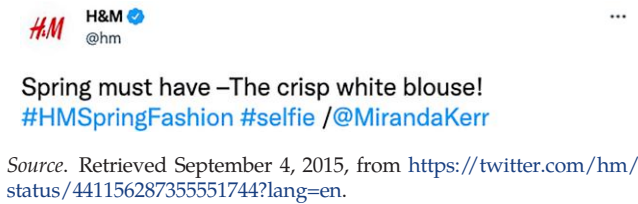
a list of the 1,308 most influential Twitter accounts in the fashion industry with an official language in English, using PeerIndex’s database and other sources from traditional fashion magazines. *PeerIndex is a leading social media analytical company that provides databases of the most influential accounts (people, brands, media, etc.)* on social media sites in a variety of industries. We obtained publicly available tweets from these most influential 1,308 tweeters posted between May 2013 and November 2017. An illustrative sample of the Twitter accounts we used is provided in Table 1.

In our color study, we complement our Twitter data set with the Google Search Volume Index (SVI) under the “Apparel” category. The Google SVI is publicly available on Google Trends and can be specified by country, time window, category, and search type. For every call, one can obtain the relative search volumes for up to six search terms. We wrote a script to make repeated calls for weekly volumes of six colors, under the “Shopping/Apparel” category, with one term fixed on a particular color, and the other five terms iterating through all colors in our study. By doing so, the fixed color serves as the benchmark, and we can obtain the relative search volumes for all colors in our study.

Extracting fine-grained information from text messages requires cognitive functions and sometimes involves subjective judgements. Ideally, every single tweet should be judged by multiple people and labeled by a majority-voting rule. However, this approach is not scalable and is therefore not applicable in our case, as there are millions of tweets that need to be processed. To overcome this challenge, we apply natural language processing (NLP) and machine learning techniques to train artificial intelligence models that automatically filter and process each tweet into meaningful numerical inputs. Typical steps for training such models include the following: (1) obtaining a training set that is already tagged and processed; (2) breaking the sentence into words or lemmas and identifying the sentence structure; and (3) using statistical tools to infer which words and sentence structures are correlated with label outcomes, thereby learning to filter and process tweets.

The training set for the models is obtained via *Amazon Mechanical Turk (AMT)*, where each tweet is manually tagged by five different workers on a series of questions. In our color study (analogous questions were used for jeans fit), the questions were as follows: (1) whether the tweet refers to color in a fashion sense; (2) which general category it refers to; (3) whether the sentiment of the message is positive, neutral, or negative; (4) whether the tweet advertises any product, and, if so, whether it is a currently available item or an item from a fashion show; (5) whether the tweet offers any type of discounts; and (6) the key words that enable them to make such judgements. A majority vote is then

Figure 3. (Color online) Example of a Twitter Account's Posted Tweet, with Specific Color Shade Term "Crisp White"



conducted to obtain the final categorization for each tweet. Consider a typical message as shown in Figure 3. In this tweet, a qualified worker will tag "relevant," "tops," "positive attitude," "yes and now," "no," with the keywords being "spring," "must have," "crisp-white," "blouse," "HMSpringFashion," "selfie," and "MirandaKerr." For each message, at least five different Turkers' inputs were recorded. We obtain the final classification by a majority-voting rule. Each such tweet is a unit of analysis in our social media data set.

The strategies that we deployed to improve the quality of classification by the Turkers are described in the online appendix. We believe our methodology for content classification has strong external validity. The classification task that we serve to the AMT Turkers in our study is simpler than the more complex tasks for which AMT-based data have been employed successfully in the literature. The existing AMT literature (Snow et al. 2008, Ipeirotis et al. 2010, Paolacci et al. 2010) shows that combining results from a few Turkers can produce data equivalent in quality to that of expert labelers for a variety of text-tagging tasks. Finally, evaluating AMT-based studies, Buhrmester et al. (2011) conclude that (1) Turkers are demographically more diverse than regular psychometric study samples, and (2) the data obtained are at least as reliable as those obtained via traditional methods, as measured by psychometric standards such as Cronbach's α , a commonly used interrater reliability measure.

4.2. Natural Language Processing

Our natural language processing task is a supervised learning problem. Denoting the input variables of tweets as X , and the labels of tweets as Y , we use a set of training data (x_i, y_i) to construct prediction rules to predict the labels $\hat{Y}$ for new tweets. The model in supervised learning used to solve such problems usually involves a mathematical structure of how to make a prediction y_i given x_i . For example, in a common linear model, the prediction is given by $\hat{y}_i = \sum_j \theta_j x_{ij}$, a linear combination of weighted input features. Models have parameters that are the undetermined part that we need to learn from the data. In linear regression problems, the parameters are the coefficients θ . In NLP tasks, we also need a way to transform textual data (i.e., a tweet) to

numerical inputs (i.e., x_i). This process is known as *feature extraction*.

Other details of how we processed tweets are provided in the online appendix. We also provide a table summarizing performance metrics for the classification process, including accuracy, which ranged from 93.2% to 99.6% for different components of tweet classification.

When fitting the forecasting models for each of the three retail partners, we use only a subset of the social media data set. This is because we work with each retail partner at different time periods, and thus only the subset available at the time a forecast is created for a retailer is fed to the models.

4.3. Retail Sales Data

Our three retailer partners include two apparel retailers (referred to as retailers A1 and A2) and one footwear retailer (referred to as retailer F). All three retailers are major players, with an average firm age of 47 years and average annual revenue of 7 billion dollars. They are all based in North America and operate worldwide. We choose to limit our scope to the women's department only. This is because, on average, women are more interested and engaged in fashion, both offline and online, than men (Freeman 2014). Women read style magazines, are active on social media platforms, and, moreover, are responsive to social media engagement. According to a marketing report conducted by Taylor (2016), "86% of women shoppers are more likely to buy a brand they've never purchased when they begin to interact with that brand on social media," and "87% are more likely to buy a brand more regularly that they check in or engage with on social media." Crum (2015) also reported that "49% of female Twitter shoppers say Twitter content has influenced their purchase decisions, which makes the platform prime real estate for brands."

A detailed overview of our retailer partners is summarized in Table 2. The three retailers have slightly different lead times, and their retail sales data were collected at different time periods. Though the time periods are not consistent across the three retail data sets, each data set spans at least two consecutive seasons and fiscal years, making cross-season validation possible. With the three data sets from different fashion retailers covering the four seasons of spring, summer, fall, and winter over a three-year period, we believe that our results and conclusions are relatively robust and have high external validity.

All three sales data sets consist of (i) transaction-level sales data, including the transaction price of each product, its corresponding sticker price, and the degree of markdown or promotion; (ii) style color characteristics, including the color attribute, subcategory, and category; and (iii) inventory information, including the

Table 2. Characteristics of the Three Retailers' Sales Data Sets

Retail Partners ^a	Apparel A1	Apparel A2	Footwear F
Initial shipment lead time	10 weeks	14 weeks	12 weeks
Retail data time periods	Spring and summer 2014, 2015	Fall and winter 2014, 2015, 2016	Fall and winter 2014, 2015
Total number of analyzed styles	1,401	2,364	2,047
Total number of analyzed style colors	4,520	8,351	4,859
Average full price (\$)	55	52	90
Average initial order quantity per style color	179	648	160
Estimated demand (full-price season) per style color	265	842	206

^aEach retailer manages a handful of brands and subsidiaries; the sales data we obtained is only a subset.

initial shipment quantity. Note that the style color level is not the SKU level because it aggregates across sizes. We do not forecast at the size level and assume that the retailers use their existing size distribution. For the scope of this project, we focus on pure colors and, therefore, drop all style colors in a pattern or mix of colors.

One common challenge in the demand forecasting literature is to recover actual demand from sales data by estimating lost sales. Sales data underestimate true demand when an item is stocked-out due to lost sales during the stock-out period. Lost sales are common among retailers, and considerable work has been done to properly estimate lost sales. Talluri and Van Ryzin (2004) provided an overview of common methods, including 1) using within-week seasonality factors to extrapolate sales based on observed sales on days when an item was known to be in-stock; and 2) using observed sales for other sizes of the same item. Garro (2011) proposed a method that averaged the result from methods (1) and (2), and we employed it on all three data sets to estimate lost sales and thereby true demand.

4.4. Final Data Set

The last step is to merge the social media data set and the retail sales data set. For the color study, matching consists of identifying which social media messages refer to the color of a particular style color carried by the retailer. We extracted 212 color terms in our social media messages, and the three retailers have 99, 143, and 84 color shades defined in their system, respectively. We used several developments in color science to accomplish color matching. The RGB system specifies a color by three numbers giving the intensity of red, green, and blue. The LCH coding developed by the Commission Internationale de l'Eclairage (CIE) defines colors numerically by the lightness, chroma, and hue. CIE also developed a metric, Delta E, for assessing the similarity of two colors defined in the LCH space. See Sharma et al. (2005) for details. Finally, Munroe (2010) conducted a massive online color survey to name over

five million colors across 222,500 user sessions. From these survey results, we obtain the RGB values for all 212 color terms extracted from social messages.

We use the RGB value to obtain corresponding LCH values and the Delta E metric to match each color in a retailer's database to the closest color mentioned in the social media messages. We had the retailers' buying teams double-check our mapping results. Two of the retailers made no adjustments, and the third made just two adjustments.

For jeans fit, a simple fuzzy string comparison was used to map both SKUs and tweets into the seven styles offered by retailer A2: Skinny, Slim, Straight, Girlfriend (GF), Boyfriend (BF), and Boot Flare. If a SKU's subcategory is "Jeans," and its style description contains one of the above style keywords, then it is classified as the corresponding style. Similarly, for a labeled tweet, if it is related to bottoms and contains one of the above fit style keywords, then it is classified as related to the corresponding style.

We merge on the time variable to align the social media features with the retail features to account for each retailer's initial shipment lead time. A summary of all the features can be found in Table 3.

5. Forecasting Methodology and Models

To evaluate the improvement in forecast accuracy from using social media information, we use each retailer's current practice forecast as the baseline and compare two constructed forecasts to it. The first constructed forecast applies statistical learning models to the same information set as was available to the retailer for their baseline forecast. The second constructed forecast applies statistical learning models to this baseline information set and social media data. We call the former the *model-only forecast* and the latter the *model-with-social-media forecast*. An improvement of our model-only forecast over the baseline would show the value of using advanced statistical tools in forecasting. An improvement of our model-with-social-media forecast over the model-only forecast indicates the value of including

Table 3. Description of Features

Feature name	Description
Non-social media features	
<i>RetailerForecast</i>	Company’s internal decision on the initial shipment quantity for each style color
<i>StickerPrice</i>	Sticker price of each SKU
<i>DemandLastSeason</i>	Estimated demand of previous season
<i>DemandLastYear</i>	Estimated demand of previous year
<i>MarketingBudget</i>	Company’s internal decision on marketing budget
<i>LagSales</i> ^a	Early weekly sales data
<i>Advertisement</i> ^a	A dummy indicating whether there are any advertisements during the week
<i>Promotion</i> ^a	A dummy indicating whether there are any promotions during the week
<i>LastAd</i> ^a	A dummy indicating whether there are any advertisements during last week
<i>LastPromotion</i> ^a	A dummy indicating whether there are any promotions during last week
<i>Season</i>	A set of dummies indicating whether it is spring, summer, fall or winter
<i>WeekOfSeason</i> dummies	A set of dummies indicating whether it is the 1st, 2nd, 3rd, ... , or 13th week of the season
<i>Holiday</i>	A dummy indicating whether there is any holiday in a week
<i>Category</i> dummy	A set of dummies indicating the category of the SKU
<i>Subcategory</i> dummy	A set of dummies indicating the sub category of the SKU
Social media features	
<i>category_tweets_count</i>	Number of tweets relevant to the category
<i>color_tweets_count</i>	Number of tweets relevant to the color
<i>category_color_tweets_count</i>	Number of tweets relevant to both the category and the color
<i>category_retweets_count</i>	Number of retweets relevant to the category
<i>color_retweets_count</i>	Number of retweets relevant to the color
<i>category_color_retweets_count</i>	Number of retweets relevant to both the category and the color
<i>category_favorites_count</i>	Number of favorites relevant to the category
<i>color_favorites_count</i>	Number of favorites relevant to the color
<i>category_color_favorites_count</i>	Number of favorites relevant to both the category and the color
<i>category_avg_sentiment</i>	Average sentiment of tweets relevant to the category
<i>color_avg_sentiment</i>	Average sentiment of tweets relevant to the color
<i>category_color_avg_sentiment</i>	Average sentiment of tweets relevant to both the category and the color
<i>photo_tweets</i>	Number of relevant tweets with photos
<i>photo_retweets</i>	Number of relevant retweets with photos
<i>photo_favorites</i>	Number of relevant favorites with photos
<i>link_tweets</i>	Number of relevant tweets with links
<i>link_retweets</i>	Number of relevant retweets with links
<i>link_favorites</i>	Number of relevant favorites with links
<i>ad_tweets_available</i>	Number of relevant tweets advertising available products
<i>ad_retweets_available</i>	Number of relevant retweets advertising available products
<i>ad_favoritets_available</i>	Number of relevant favorites advertising available products
<i>ad_tweets_fashion_show</i>	Number of relevant tweets advertising products appeared in fashion shows
<i>ad_retweets_fahion_show</i>	Number of relevant retweets advertising products appeared in fashion shows
<i>ad_favorites_fashion_show</i>	Number of relevant favorites advertising products appeared in fashion shows
<i>discount_tweets</i>	Number of relevant tweets with discounts
<i>discount_retweets</i>	Number of relevant retweets with discounts
<i>discount_favorites</i>	Number of relevant favorites with discounts
<i>SVI</i>	Google Search Volume Index

^aFeatures only available for in-season forecasting.

social media data in the information set. Finally, an improvement of our model-with-social-media forecast over the baseline gives the total value of having both advanced models and social media information. We describe the details of the baseline and the two constructed forecasts in the following.

Our goal is to predict full-price season demand at the style color or jeans fit level, which can be used by the retailers as an input to set the initial shipment quantity. However, since we have much more granular data on hand, the approach we take that maximizes statistical power is to predict weekly sales for the full-price period (usually 8 to 12 weeks in duration), and then

aggregate the predictions to get the predicted total quantity. This approach leverages the granularity of our data set to generate many more data points (one data point per style color week) than simply predicting the total quantity (one data point per style color). Moreover, in the aggregation process, prediction errors can cancel each other out to some extent.

5.1. Current Performance

All three retailers had an existing process for creating a forecast of what they expected to sell of each item in a coming season. We use these forecasts as the baseline against which to measure improvement due to our

proposed methodology. None of the current practices takes social media information into consideration, nor did they implement state-of-the-art machine learning techniques in their forecasting models. In general, current practice considers historical data, season over season and year over year trends for each category, and planned marketing budgets.

5.2. Baseline Information Forecast Model

In the baseline model without social media information, we forecast the total demand of a product color or jeans fit style during its full-price life cycle as a function of all the baseline information that the retailer has on hand when making the initial buy decision:

$$Demand_{it} = f^{baseline}(Demand_{it}^{season-1}, Demand_{it}^{year-1}, RetailerForecast_{it}, StickerPrice_i, X_{it}) + \epsilon_{it} \quad (1)$$

$$Demand_i = \sum_t Demand_{it}, \quad (2)$$

where $season - 1$ represents last season, $year - 1$ represents last year, X_{it} are the characteristics specific to the style color item i (e.g., color, category fixed effect) and its selling season t (e.g., seasonal fixed effect), and ϵ_i is the idiosyncratic demand shock. $RetailerForecast_{it}$ is the quantity the retailer initially forecast, without our analysis, that style color i would sell during the coming full-price season. $StickerPrice_i$ is the initial price of item i set by the retailer prior to the start of the sales season. In the process we are proposing, this initial shipment decision would be updated based on the more accurate forecasts derived from social media. The model features are listed in Table 3. For new styles that do not have historical data, demand for the last season and year is approximated by the category average. We use a general function form $f(\cdot)$ because it can take many forms, as specified by the three forecasting models we apply.

5.3. Full Information Forecast Model

In the models that include social media data (i.e., using the full information set as its input), we have the same structure as the baseline model, with the addition of social media variables:

$$Demand_{it} = f^{full}(Demand_{it}^{season-1}, Demand_{it}^{year-1}, RetailerForecast_{it}, StickerPrice_i, Media_{i,t-l}, X_{it}) + \epsilon_{it}, \quad (3)$$

$$Demand_i = \sum_t Demand_{it}, \quad (4)$$

where $Media_{i,t-l}$ represents all lagged social media features related to the style color item, aggregating over the item's expected normal life cycle. The social media features are described in Table 3. The variable *Number*

of *Relevant Tweets with Links* tabulates instances of links (same for retweets and favorites). Typically, these are links to a store page, an ongoing live fashion show, or the image attached to the tweet.

5.4. Fitting and Evaluation

We use out-of-sample forecasting accuracy as our evaluation metric. Recall that each data set spans at least two consecutive seasons and fiscal years. Therefore, we could fit all forecasting models over the first year and evaluate on the second year. An alternative is to use 10-fold cross-validations, where a model is fit over nine-tenths of the data and evaluated on the left-out one-tenth of the data, and the process is repeated 10 times. We adopt the first approach so that we mimic the actual forecasting practices taken by fashion retailers.

Denote by T^F and T^E the fitting time period and evaluating time period for all retailers. Let $f^{baseline}(T^F)$ and $f^{full}(T^F)$ denote the baseline and full information models trained on the fitting data. When we construct forecasts for the out-of-sample data, we apply the estimated parameters and hyperparameters from the two trained models directly to the out-of-sample data, namely, the data set during time period T^E . None of the models is reestimated for the out-of-sample period so that we are guaranteed a true out-of-sample forecast. Note that the dependent variable we are predicting here is the total full-price season demand of each style color, which is an aggregate-level quantity.

5.5. Three Forecasting Models

We fit our data sets with three commonly used forecasting models, a lasso regression model with regularization, a support vector machine model with RBF kernel, and a random forest model (see Hastie et al. 2009 for descriptions). The random forest model produced the lowest out-of-sample MAD (mean absolute deviation) for all three retailers, and so we focus exclusively on this model in the results below.

There is another popular class of forecasting model, the neural networks. However, we do not consider neural network models here because they are known to perform poorly on small samples with high-dimensional features since they require enormous sample size to avoid overfitting. Our sample size is too small to satisfy this requirement.

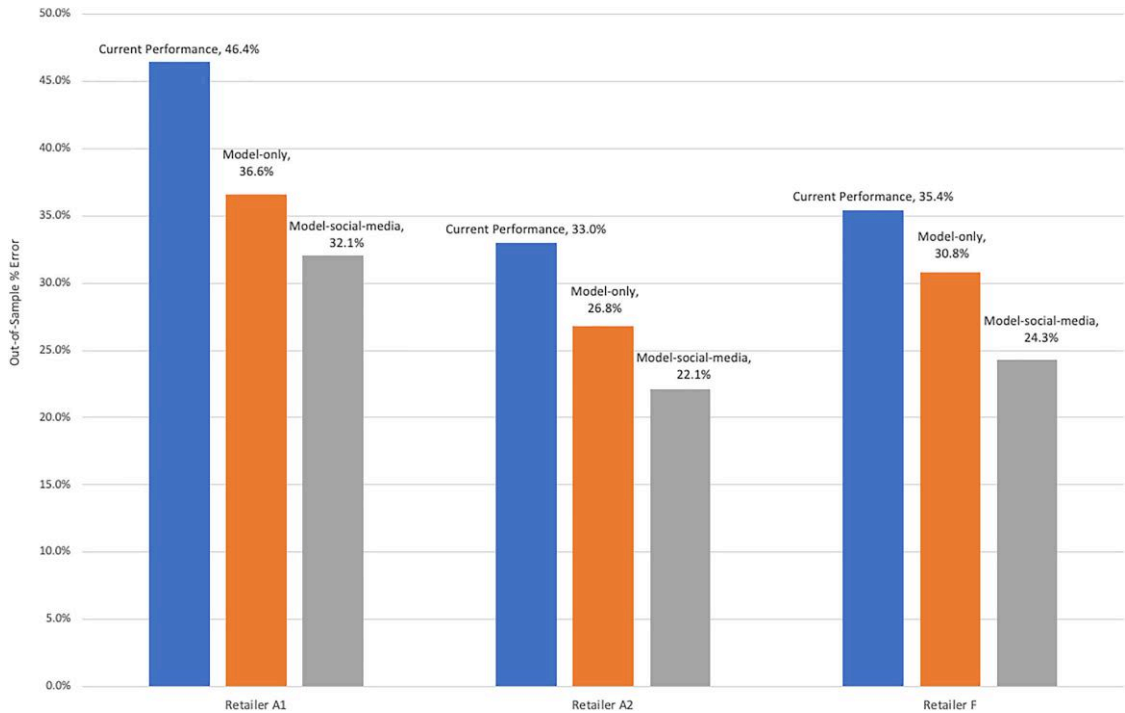
6. Results

In this section, we compare the forecast error of the processes we have developed to the retailer's current process.

6.1. Color

Figure 4 shows out-of-sample MAD as a percentage of total demand (computed as the absolute difference between the forecast and demand for each style color,

Figure 4. (Color online) Out-of-Sample MAD as a Percentage of Total Sales for Retailer A1 (left), A2 (center), and F (left) for the Random Forest Model and Color Analysis



summed over all style colors and divided by total sales) for the three forecasts and three retailers.

All statistical model-only forecasts are more accurate than the retailers’ current performance, and incorporating social media data into the statistical models further reduces error. Overall, our process reduces error for the retailers by 28% (i.e., $1 - 33.6/46.4$), 24%, and 26%, respectively, for the three retailers. The consistent results across

Figure 5. (Color online) Comparison of Five Subsets of Social Media Information Across Three Retailers, MAD as a Percentage of Total Sales, Color Analysis

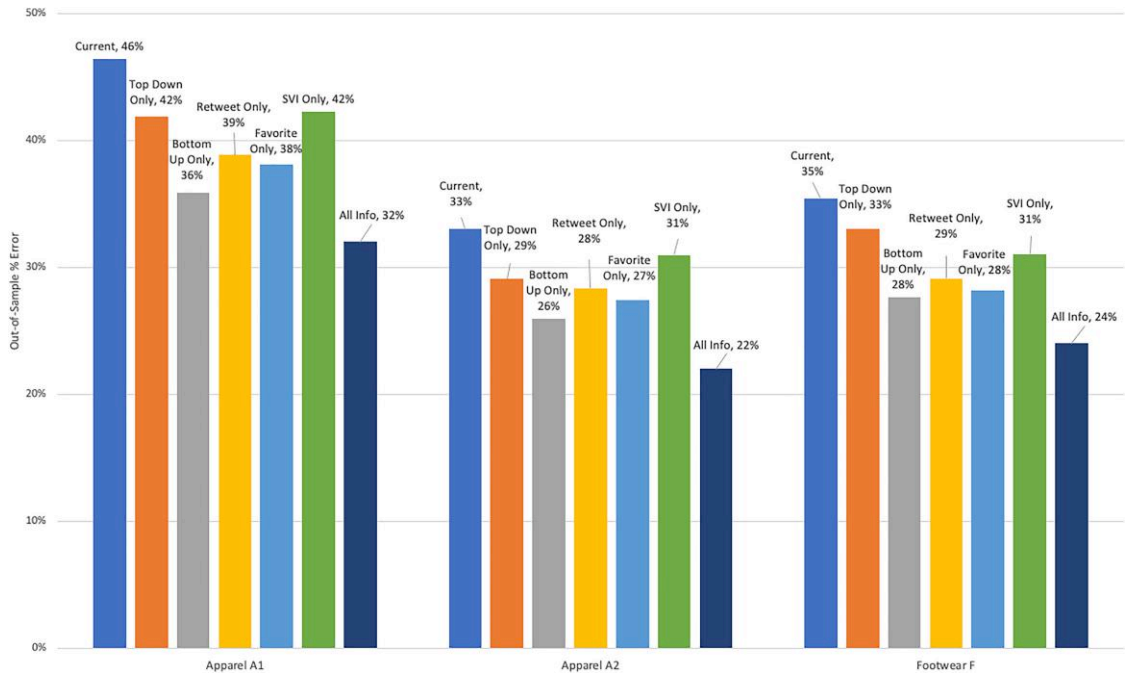
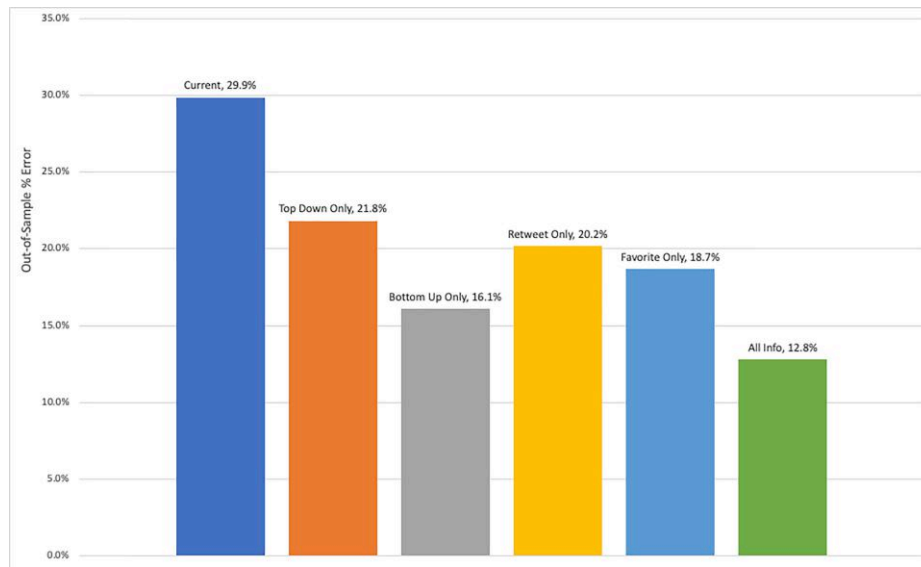


Figure 6. (Color online) Comparison of Five Subsets of Social Media Information for Retailer A2, MAD as a Percentage of Total Sales, Jeans Fit Analysis

all three retailers offer a strong validation to our key message: **publicly available social media information adds value in the challenging task of predicting fashion trends.**

In addition to using tweets, retweets, favorites, and Google SVI altogether for predictions, we explored the value of different subsets of the social media information. We examine the predictive value of five different subsets: (1) top-down fashion industry messages only (i.e., tweets); (2) bottom-up consumer preference indicators only (i.e., retweets, favorites, and SVI); (3) retweets only; (4) favorites only; and (5) SVI only. We replicate the five predictions for all three retail partners and present the results in Figure 5.

We find that the consumer preference indicator information is the most valuable of the five, almost as powerful as including everything. In general, subsets of Twitter information perform better than Google SVI, suggesting that information from Twitter, after being properly processed by NLP, is more fine-grained and valuable than SVI. The patterns are mostly consistent across all three retailers.

Comparing the relative contributions of top-down and bottom-up messages for forecast accuracy improvement sheds light on the important question of whether fashion change is driven by a push from the fashion industry on changes they try to dictate, or by changes in consumer preferences. If we compare current performance to all info, we see a reduction in MAD of 14%, 11%, and 11%, respectively, for the three retailers, for an average reduction of 12%. Comparing current to bottom-up only, we see improvements of 10%, 7%, and 7%, respectively, for an average improvement of 8%. Thus, we see that $8/12 = 67\%$ of the improvement comes

from the bottom up, and only 33% from the top down. This suggests that fashion changes are not mostly dictated by the fashion industry but rather come more from changes in consumer preferences.

6.2. Jeans Fit

For jeans, the unit of analysis is just style, aggregated across all colors. If we further break them down to the style color level, then there would be too few samples per style color cohort. The analysis on jeans is almost exactly the same as for color, except that three covariates were removed, because all colors are aggregated. In Table 3, for the color analysis, there are two features derived from the count of tweets: `category_tweets_count` and `category_color_tweets_count` (same for retweets and favorites). For the jeans analysis, the two features reduced to a single feature: `style_tweets_count` (same for retweets and favorites).

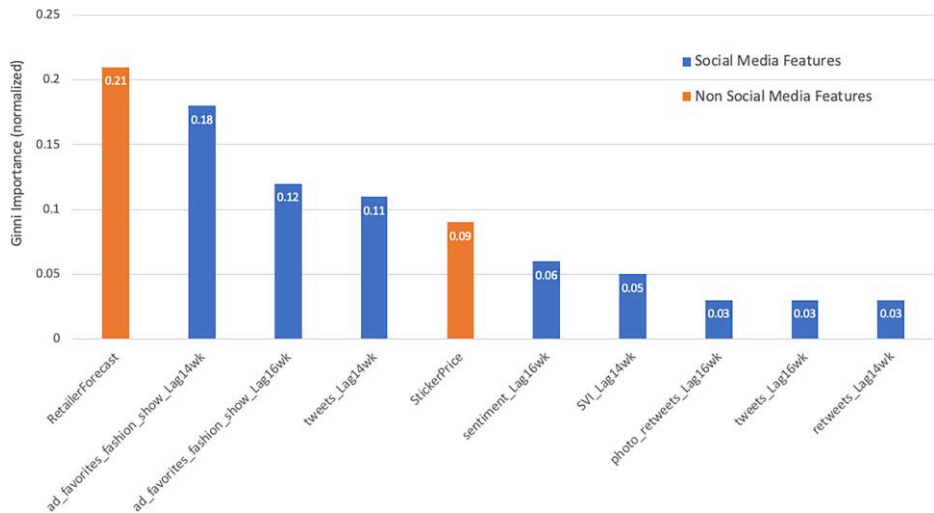
For the jean analysis, random forest again performed the best. Figure 6 gives MAD as a percentage of total sales for five subsets of social media information. As with color, the All info forecast significantly reduces error, from 29.9% to 12.8%. And again, most of this improvement comes from bottom-up only, which contributes $(29.9 - 16.1)/(29.9 - 12.8) = 81\%$ of the improvement.

7. Analysis of Features, Lead Time, and Markets

7.1. Features

Figure 7 shows the top 10 features of the random forest model for a 14-week lead time for the color study and Figure 8 for the jeans fit study. Results from models with other lead times have similar patterns. The fact

Figure 7. (Color online) Top 10 Features of the Random Forest Model with Lead Time of 14 Weeks, Color Analysis



that in the color study eight out of the top 10 features are social media lends support to the value of social media. Five of the eight social media features are bottom-up, which lends support to the conclusion that fashion changes come more from consumers than from the fashion industry. Similarly, for the jeans fit study, eight out of the top 10 features relate to social media, and seven of those eight are bottom-up, lending additional evidence that fashion change comes from the bottom up much more than from the top down

7.2. Prediction Horizons

We repeat the forecasting process for our color analysis with other prediction horizons, ranging from two weeks to six months, as well as an in-season forecast (i.e., no lead time) to test the robustness of our results and explore potential heterogeneity on the time dimension. The analysis of varying prediction horizons for jeans fit produced results basically the same as for color and is

omitted here due to space constraints. This range of lead times covers both ends of the spectrum—players such as department stores with a six-month lead time, and fast-fashion retailers with made-to-order supply chains.

An in-season forecast is different from a preseason forecast in that early sales data are available for the former and are proven to be significant in improving forecast accuracy (Fisher and Raman 1996). The main purpose of our forecasting comparison is to understand the predictive power of social media across lead times. Therefore, we first present the comparison results where the information set used for prediction is consistent across all lead times and early sales data are not used for the in-season forecast. We then examine the in-season forecast alone to compare results when early sales data are included.

In this analysis, we pool all three data sets to maximize the statistical power and only report the results of the best-performing random forest model. Since the earlier forecasting results are consistent across all three

Figure 8. (Color online) Top 10 Features of the Random Forest Model with Lead Time 14 Weeks, Retailer A2, Jeans Fit Analysis

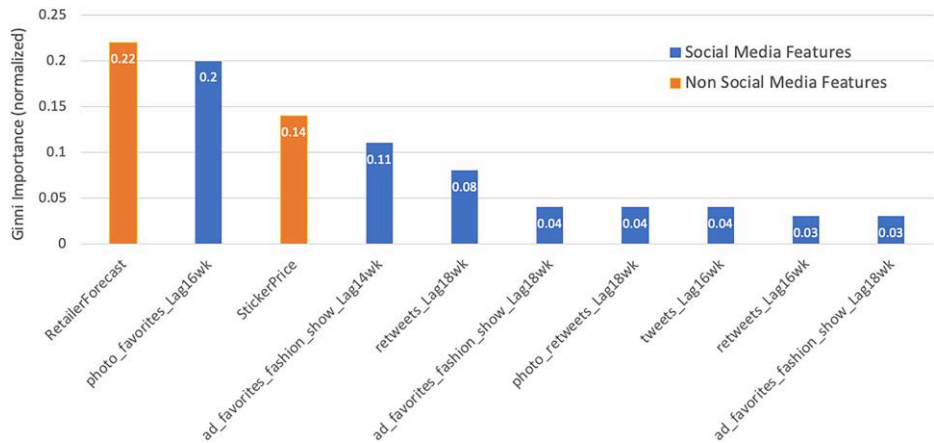
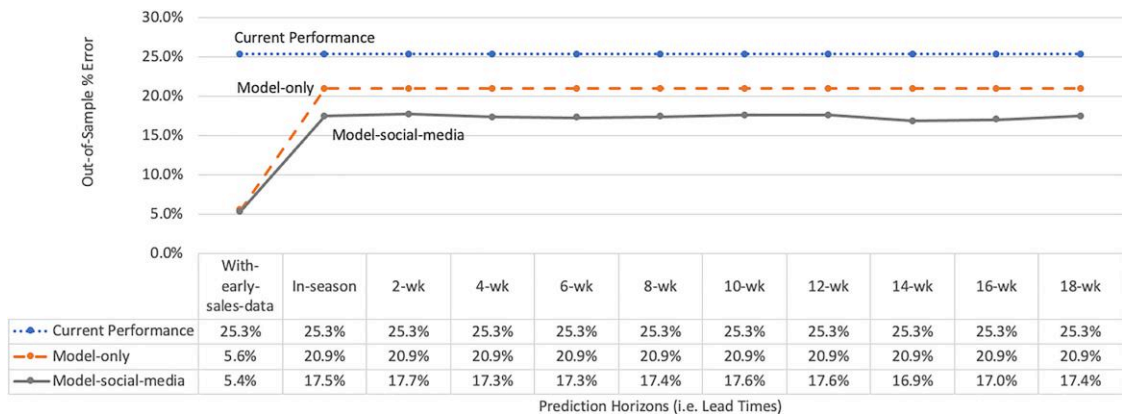


Figure 9. (Color online) Out-of-Sample MAD as a Percentage of Total Sales of Random Forest Model and Social Media Information Across Prediction Horizons (i.e., Lead Times), Color Analysis



data sets, pooling them together should not alter any results and improves statistical power. To examine the value of social media information across forecasting horizons, we plot in Figure 9 the out-of-sample MAD of the three scenarios—baseline, random forest model only, and model plus social media features—over varying lead times.

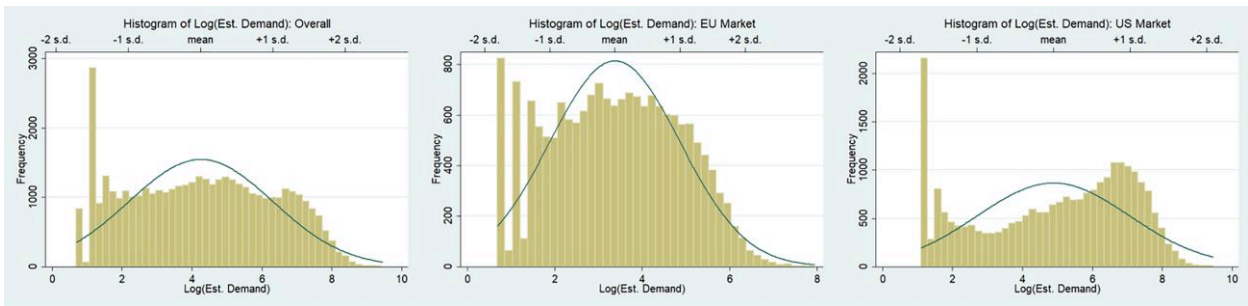
We see that the random forest model and social media information each individually improve the forecasting results significantly. The results indicate that the value of social media data is consistent and robust across all lead times. The six-month-long effective forecasting horizon may be surprising, but it coincides well with the fashion industry timeline—from the “Big Four” fashion shows all the way to the targeted season. Note that this does not suggest that fashion influencers are always tweeting about the future targeted season during this six-month-long period. They could well be buzzing about the current season, but with our natural language processing and machine learning techniques, the forecasting model is able to extract relevant information and make accurate predictions. We can observe in Figure 9 that the forecast lead time has virtually no impact on forecast accuracy.

We repeat the in-season forecasting process, this time including available early sales data in both the model-only and model-social-media setups, and compare their out-of-sample MAD. The predictive value of social media when early sales data are available, not surprisingly, is only marginal and not statistically significant. With early sales data, the model alone has a MAD of 5.6% of total sales, and adding social media info only brings MAD down to 5.4% of total sales.

7.3. Markets

The retail sales data provided by apparel retailer A2 includes their EU and U.S. markets, which enables us to examine the heterogeneity across the two regions for our color analysis. Figure 10 shows histograms with fitted normal curve of the log of estimated demand (sales corrected for lost sales, as described in Section 4.3) for the retailer’s overall EU and U.S. markets, respectively. We clearly see two significantly different demand patterns. The estimated demand is relatively normal in the left (overall) and middle (EU market) panel, whereas the estimated demand is highly skewed to the right in the U.S. market. All three histograms have inflated low values, which is common with weekly demand. For this

Figure 10. (Color online) Histograms of Log of Estimated Demand of Apparel Retailer A2: Overall (left), EU Market, and U.S. Market (right), Color Analysis



retailer, out-of-sample MAD as a percentage of total sales for current performance, statistical model only, and statistical model plus social media was 24.8%, 19.1%, and 17.1%, respectively, for the EU market, and 40.8%, 36.8%, and 28.9%, respectively, for the U.S. market. The total reduction in MAD for the EU market is 31% ($1 - 17.1/24.8$) and for the U.S. market 29%. The results display an interesting pattern. Although the total improvement in MAD in the two markets is close, the forecasting model contributes significantly more than social media features in the EU market, and vice versa for the U.S. market. Intuitively, normally distributed demand in the EU market is easier to predict using a statistical model alone, and social media features are simply icing on the cake. To the contrary, predicting the right-skewed demand in the U.S. market using a model alone does not achieve the best performance, and adding social media features brings significant value.

8. Hypotheses Validation and Discussion of Why Social Media Is Predictive

We find support for Hypotheses 1 and 2. Figures 4, 5, and 6 show that social media data significantly improve forecast accuracy, and most of this impact comes from retweets/favorites and Google SVI, the bottom-up data that are indicative of consumer sentiment. Interestingly, Hypothesis 3 is not supported by our analysis. Figure 9 shows that forecast accuracy does not improve as we get closer to the season. This suggests that efforts to reduce product development lead time may not provide a benefit. However, once we are within the season, early sales significantly improve forecast accuracy, supporting the business model of firms like Zara that shorten their lead time by so much that they can react within a season.

As with all of the prior efforts using social media to improve demand forecasts that were reviewed in Section 2, our primary purpose here is to predict, not to explain, a distinction well described in Shmueli (2010). That said, it is still useful to consider the potential mechanisms that might explain social media being predictive. We envision four reasons for our findings: (1) tweets from an apparel supplier are a form of online advertising; (2) retweets are a form of social networking in which the retweeter influences their connected neighbors; (3) SVI is predictive because product research using Google is often a step prior to purchase; and (4) retweets, favorites, and SVI act as a poll of consumers by which they signal their color preferences.

The first three of these align with prior findings in the literature. Crum (2015) discusses tweets as advertising, and Cui et al. (2017) cite the creation of awareness as a potential mechanism. Goel and Goldstein (2013), Trusov et al. (2009), Goldenberg et al. (2012), Centola (2010), Christakis and Fowler (2007), and Cui et al. (2017) have all discussed social networking as an

influencer of behavior. Finally, Geva et al. (2017), Vosen and Schmidt (2011), and Wu and Brynjolfsson (2009) have also found SVI to be predictive.

Figures 5 and 6 provide insight as to the relative importance of these effects by giving the errors for four pure signals: (1) tweets (advertising from the brands); (2) retweets; (3) favorites; and (4) SVI. Of these, favorites had the lowest forecast error, suggesting that consumer polling is the biggest effect. Retweets had the second lowest error, suggesting that social networking was the second largest effect. Interestingly, the tweets themselves were the least predictive, indicating that social media as a form of online advertising is the least effective of the four mechanisms, other than that the tweets engender an online “conversation” among consumers, which is impactful.

To better understand these findings, we sent our paper for comment to a group of industry experts that included chief operating officers of fashion apparel and footwear companies, heads of apparel and fashion industry research groups, heads of companies that did fashion forecasting, and an academic specializing in social media. The list of these individuals and their comments is included in the online appendix. Here, we summarize some of their key points.

They agreed and it aligned with their own experience that social media would be predictive and that bottom-up signals would be more predictive than top-down advertising from brands. Some typical comments included the following: “The genuineness of actual consumer media posts (as compared to posts from the retailer itself) is the difference to me here.” And: “A generation ago fashion trends are top-down. There were powerful and influential designers who dictated trends using their fashion shows, product lines, and fashion magazine photos. Since the late 1990s fashion trends seem to be more peer to peer, social, and bottom-up. Think of people like Coco Chanel, Christian Dior, Armani . . . There are no modern equivalents.”

The other interesting point made was that bold new colors are exciting and draw people into the store, but they do not sell as well as their hype, as emphasized in the following comment: “Further, it’s often true that the more fashion-forward colors (e.g., something a bit on the wilder side vs. more neutral, boring colors) are the colors that a retailer would promote, either in the store front windows or—these days—in their social media posts. These colors are more interesting and attract attention, but the consumer ultimately buys the “boring” color when their practical side kicks in.” This idea was seconded by another expert who independently commented: “You have to be careful for predicting unusual color/style. When something very new/fresh comes out, people tend to over-react. In the course of my career, I had made these mistakes. (They did not

sell as I expected. When it comes to real purchase, they question twice.)”

As to where social media is predictive, we agree with the comment, “In general, social media can predict sales for products people discuss on social media. So I don’t think sales of refrigerators could be predicted with social media.”

9. Discussion and Conclusions

We have conducted a large-scale study involving three retail partners, about 500,000 tweets analyzed, and multiple years. Our analysis shows that incorporating fine-grained social media information has significant predictive power in forecasting color and jeans fit demand occurring months later. This can help fashion retailers make their initial shipment quantity decision, which is difficult, due to long manufacturing lead times, and the elusive nature of fashion demand. The forecasting accuracy improvements hold for out-of-sample predictions for all three retailers, in both apparel and footwear markets, across two geographic regions (European Union and United States), and over a wide range of lead times (from two weeks to six months). We thus believe our results have external validity and broad applicability.

For fashion retailers, making better initial shipment quantity decisions leads directly to substantial operational benefits, most importantly, fewer lost sales due to stock-outs and reduced markdown expense due to less excess inventory. If one is interested in forecasting at different levels of detail from our setting, such as regional or store level sales, the same framework still applies. It only requires breaking down both sales and social media data by locations.

We also test the forecasting value of social media information for in-season forecasts when sales data are available. Not surprisingly, once actual sales are available, they are a better demand signal than social media.

Our conversations with our partner retailers and several industry practitioners at leading fashion forecast companies confirm the important practical implications of our results. The current forecasting process at those companies is highly artistic, relying heavily on experienced fashion experts’ qualitative inputs. As a matter of fact, social media information is one of the sources these experts say they keep an eye on to obtain implications and draw conclusions. Therefore, this study provides a methodology for them to quantitatively use social media information to improve fashion demand forecasting systematically. After we shared our results, one of the retailers is actively planning to adopt our method.

We conclude with a summary of insights that follow from our research.

(1) Using social media significantly improves forecast accuracy.

(2) Fashion firms, seeking to dictate what will be available for the next season, choose colors they wish to push in an upcoming season, and then both “advertise” them via Twitter and supply inventory in those colors. However, the fact that fashion insiders are trying to dictate colors on the supply side does not mean customers are simply going to buy those colors; otherwise, we would not see the high markdowns and stock-out prevalent in the fashion industry. In fact, our analysis showed that the components of social media that capture consumer sentiment (retweets, favorites, and SVI) account for 67% of the color forecast improvement and 81% of the jeans fit improvement, suggesting that fashion changes are more driven by consumer taste changes than industry push. One way to view this is that social media are more than new channels via which fashion firms can broadcast their message to consumers. Rather, it turns that broadcast into a “conversation” in which consumers can express what they feel about fashion via retweets and favorites. Our research suggests that fashion firms would be well advised to “listen” to what their customers are saying in that conversation.

(3) This aligns with an industry shift that has variously been called the “death of the merchant prince” and “democratization of fashion.”

(4) This is further supported by feature analysis showing that seven of the top 10 features for both color and jeans are retweets, favorites, and Google search, all expressions of customer preferences, and by comments from industry experts.

(5) One reason tweets from brands are less predictive is that they tend to push trendy new styles that the brands think capture consumer attention, while at time of purchase consumers tilt toward safer more traditional styles.

(6) Despite the effort of brands to reduce lead time, we found that forecasts made close to the sale season were not more accurate than those made as much as 18 weeks before the sales season. However, having actual sales significantly improves forecast accuracy.

(7) European sales are normally distributed, whereas U.S. sales are skewed to the right. Moreover, the prediction value of social media is much greater for the United States than for Europe.

As literature on social media continues to proliferate in fields like economics and information systems, there is a solid foundation and huge potential for researchers to explore these data in novel ways that make firms operate more efficiently. We hope this study contributes to the usage of social media data in tackling operational problems and inspires others to conduct further studies. Some opportunities for additional research include use of other social media, such as Instagram, and using social media to forecast things people like to discuss online, such as sales of other “fashion” products like video games, sale of movies tickets, and cryptocurrency prices.

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